Question 14: Hive DML commands

Aim

To perform Data Manipulation Language (DML) operations in Hive, such as loading, inserting, selecting and exporting

table data.

Procedure

1. Create or select a Hive database and create a table.

2. Load records from a local file or HDFS into the table.

3. Insert additional records using INSERT INTO or overwrite existing table data using INSERT OVERWRITE.

4. Retrieve records using SELECT with filtering and sorting clauses.

5. Export table data to a local path or HDFS location when required.

Program / HiveQL commands

```
CREATE DATABASE IF NOT EXISTS college;

USE college;

CREATE TABLE student (

sid INT,

sname STRING,

department STRING

)

ROW FORMAT DELIMITED

FIELDS TERMINATED BY ',';

-- Load a local file into the table

LOAD DATA LOCAL INPATH '/home/student/student.txt'

INTO TABLE student;

-- Insert a new record

INSERT INTO TABLE student

VALUES (104, 'Kumar', 'CSE');

-- Replace table contents with the query result

INSERT OVERWRITE TABLE student

SELECT sid, sname, department

FROM student
```

```
WHERE department = 'CSE';

-- Retrieve all records

SELECT * FROM student;

-- Retrieve selected records

SELECT sname, department

FROM student

WHERE sid >= 102;

-- Export query results to HDFS

INSERT OVERWRITE DIRECTORY '/output/student'

ROW FORMAT DELIMITED

FIELDS TERMINATED BY ','

SELECT * FROM student;
```

Output of Program

The LOAD and INSERT statements complete successfully.

```
SELECT * FROM student;

104 Kumar CSE
```

The export operation writes the selected rows to /output/student in HDFS.

Result

Hive DML commands were used to load, insert, query and export records from the student table.

The INSERT OVERWRITE example replaces the table contents with the filtered query result. Run it only when replacement is

intended.